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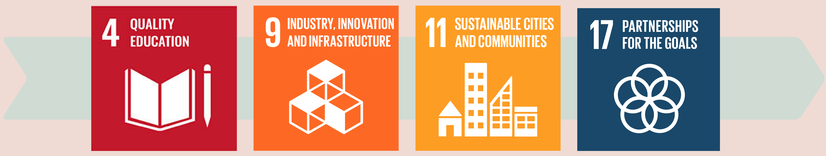
This InfoVis project is for submission to INFOSCI 301 Data Visualization and Information Aesthetics Spring 2026, instructed by Professor Luyao Zhang at Duke Kunshan University. We thank our professor, peer classmates, and community partners , including the Jiading District Urban Planning Exhibition Hall, the Fengxian Waving Cube Sci-Fi Immersive Museum, and the Kunshan Museum of Chinese Opera Genres (Kunqu Opera Living Heritage) , for their contributions to this project. We also extend our gratitude to industry contributors for their valuable insights.

A DATA-DRIVEN URBAN PLANNING VISUALIZATION FRAMEWORK FOR JIADING DISTRICT

This project transforms urban planning exhibitions from aesthetic showcases into transparent, data-driven storytelling tools. By creating an interactive visualization framework, it traces how data leads to policy decisions and urban outcomes. The framework targets both policymakers and the public, inviting users to explore policy impacts, fostering civic engagement and transparency.

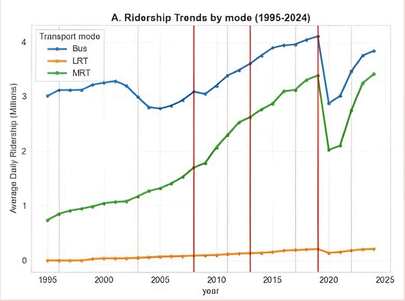
SDG

This project advances multiple Sustainable Development Goals. It champions SDG 11 through inclusive urban planning, enabled by SDG 17 multi-stakeholder partnerships. The work is powered by SDG 9 civic tech innovation and promotes SDG 4 via community-based public education on sustainable cities.



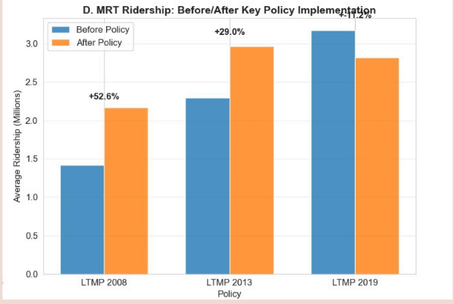
This project serves three interconnected communities. First, the residents, who are directly impacted by planning decisions but lack insight into them. Second, policy makers and district leaders, who need transparent, data-driven tools to communicate policies and justify decisions. Third, university students in planning and data science, who gain practical experience developing civic technology for real community needs.

DATA 1



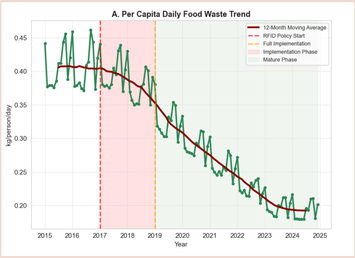
This graph tracks ridership trends from 1995 to 2024, highlighting the dominance of Buses and the rapid expansion of the MRT. In contrast, LRT usage has remained stagnant with minimal growth. Following a sharp dip in 2020, both Bus and MRT have shown a robust recovery toward their historical peaks.

DATA 2



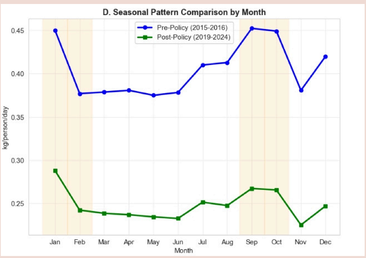
This chart illustrates the impact of Singapore's Land Transport Master Plans (LTMP) on MRT ridership. While the 2008 (+52.6%) and 2013 (+29.0%) policies saw massive growth, the 2019 implementation coincided with an 11.2% decline, likely reflecting the unique disruptions to global mobility during that period.

DATA 3



This chart tracks South Korea's per capita food waste from 2015 to 2024. Following the 2019 full implementation, the 12-month moving average shows a sustained and significant decline in daily waste generation.

DATA 4



This chart shows South Korea's seasonal waste trends, comparing per-capita output before and after policy implementation, proving the success of the policy.

POLICY IMPACT

GYEONGGI-DO :

- Pre-policy average (2015-2016): 0.405 kg/person/day
- Post-policy average (2019-2024): 0.248 kg/person/day
- Overall reduction: 38.7%
- Annual reduction rate: 6.5%/year

SINGAPORE:

- 3.2M daily MRT riders
- 75% peak-hour public transport trips
- Congestion cost savings: -$2.1B/year
- CO_2 transport emissions ↓ 18%

DATA PROCESSING, ANALYSIS, AND VISUALIZATION TOOLS

Data Collection and Curation
Data cleaning, transformation, and multi-dataset integration using Python with Pandas & NumPy. Supports temporal analysis and aggregation.

Data Processing and Governance
Ensured through Jupyter Notebooks, providing reproducible analysis workflows with embedded documentation and code for transparency in methodology.

Interactive Exploration Interface
Naive interactive application created using Steamlit.

Insight Generation
Derived from initial visualizations and statistical analysis embedded in Jupyter Notebooks, created using seaborn and matplotlib.